



Lakshya JEE 2.0 (2024)

Differential Equation

DPP-03

- Solve: $(y^2 - 2xy)dx = (x^2 - 2xy)dy$.
 - $-\frac{1}{3}\log\left|\frac{y}{x}\left(\frac{y}{x}-1\right)\right| = \log|cx|$
 - $-\frac{1}{3}\log\left|\frac{x}{y}\left(\frac{x}{y}-1\right)\right| = \log|cx|$
 - $-\frac{1}{3}\log\left|\frac{y}{x}\left(\frac{y}{x}+1\right)\right| = \log|cx|$
 - $-\frac{1}{3}\log\left|\frac{x}{y}\left(\frac{x}{y}+1\right)\right| = \log|cx|$
- If $\frac{dy}{dx} = \frac{y + x \tan y/x}{x}$, then $\sin \frac{y}{x}$ is equal to
 - Cx^2
 - Cx
 - Cx^3
 - Cx^4
- The solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x}$ satisfying the condition $y(1) = 1$ is
 - $y = \ln x + x$
 - $y = x \ln x + x^2$
 - $y = xe^{(x-1)}$
 - $y = x \ln x + x$
- Identify the statement (s) which is/are true?
 - $f(x,y) = e^{y/x} + \tan \frac{y}{x}$ is homogeneous of degree zero.
 - $x \cdot \log \frac{y}{x} dx + \frac{y^2}{x} \sin^{-1} \frac{y}{x} dy = 0$ is homogeneous differential equation.
 - $f(x,y) = x^2 + \sin x \cdot \cos y$ is not homogeneous.
 - $(x^2 + y^2)dx - (xy^2 - y^3) dy = 0$ is a homogeneous differential equation.
- The general solution of the differential equation $[2\sqrt{xy} - x]dy + y dx = 0$ is (Here $x, y > 0$)
 - $\log x + \sqrt{\frac{y}{x}} = c$
 - $\log y - \sqrt{\frac{x}{y}} = c$
 - $\log y + \sqrt{\frac{x}{y}} = c$
 - None of these
- The slope of tangent at (x, y) to a curve passing through $(2, 1)$ is $\frac{x^2 + y^2}{2xy}$, then the equation of the curve is
 - $x(x^2 + y^2) = 10$
 - $x(x^2 - y^2) = 6$
 - $2(x^2 - y^2) = 6y$
 - $2(x^2 - y^2) = 3x$
- If the solution of the differential equation $\frac{dy}{dx} = \frac{x^3 + xy^2}{y^3 - yx^2}$ is $y^k - x^k = 2x^2y^2 + \lambda$ (where, λ is an arbitrary constant), then the value of k is
 - 2
 - 4
 - 1
 - $\frac{3}{2}$
- If $(y^3 - 2x^2y)dx + (2xy^2 - x^3) dy = 0$, then the value of $xy \sqrt{y^2 - x^2}$, is
 - $y^2 + x$
 - xy^2
 - any constant
 - None of these

9. A curve passes through the point $\left(1, \frac{\pi}{6}\right)$. Let the slope of the curve at each point (x, y) be $\frac{y}{x} + \sec\left(\frac{y}{x}\right)$, $x > 0$. Then the equation of the curve is
- $\sin\left(\frac{y}{x}\right) = \log x + \frac{1}{2}$
 - $\operatorname{cosec}\left(\frac{y}{x}\right) = \log x + 2$
 - $\sec\left(\frac{2y}{x}\right) = \log x + 2$
 - $\cos\left(\frac{2y}{x}\right) = \log x + \frac{1}{2}$
10. Solution of the differential equation $x \frac{dy}{dx} = y + \sqrt{x^2 + y^2}$,
- $x + \sqrt{x^2 + y^2} = C y^2$
 - $y + \sqrt{x^2 + y^2} = C y^2$
 - $x + \sqrt{x^2 + y^2} = C x^2$
 - $y + \sqrt{x^2 + y^2} = C x^2$
11. $(x^2 + y^2)dy = xy dx$. If $y(x_0) = e$, $y(1) = 1$, then value of $x_0 =$
- $\sqrt{3}e$
 - $\sqrt{e^2 - \frac{1}{2}}$
 - $\sqrt{\frac{e^2 - 1}{2}}$
 - $\sqrt{\frac{e^2 + 1}{2}}$
12. Solve: $x^2 \frac{dy}{dx} - 3xy = 4x^4 + 2x^2$
- $\frac{y}{x^3} = \ln|x| + C$
 - $y = x^2 \ln|x| - \frac{1}{x} + C$
 - $y = x \ln|x| - \frac{1}{x^2} + C$
 - $\frac{y}{x^3} = 4 \ln|x| - \frac{1}{x^2} + C$
13. Solve: $ydx - xdy + \ln x dx = 0$. Choose from given options
- $y = -(\log x + 1) + cx$
 - $y = e^x + cx$
 - $y = x^{\ln x} + cx$
 - $y = -\log x^2 + cx$
14. Solve: $\frac{dy}{dx} = y \tan x - \sin x$.
- $y \cos x = \cos 2x + C$
 - $y \cos x = \frac{\cos 2x}{4} + C$
 - $y \sin x = \cos 2x + C$
 - $y \sin x = \frac{\cos 2x}{4} + C$
15. The solution of differential equation $(x^2 - 1) \frac{dy}{dx} + 2xy = \frac{1}{x^2 - 1}$ is:
- $y(x^2 - 1) = \frac{1}{2} \log \left| \frac{x-1}{x+1} \right| + c$
 - $y(x^2 + 1) = \frac{1}{2} \log \left| \frac{x-1}{x+1} \right| - c$
 - $y(x^2 - 1) = \frac{5}{2} \log \left| \frac{x-1}{x+1} \right| + c$
 - None of these
16. Solve: $(x + 3y + 2) \frac{dy}{dx} = 1$. Considering linear differential equation
- $x = -3y - 5 + Ce^y$
 - $y = -3(y+1) + Ce^x$
 - $x = e^x + (x+1)C$
 - $y = C.e^x + (x+1)$
17. The integrating factor of the differential equation $(1 + x^2) \frac{dy}{dx} + y = e^{\tan^{-1} x}$ is
- $\tan^{-1} x$
 - $1 + x^2$
 - $e^{\tan^{-1} x}$
 - $\log_e (1 + x^2)$

18. If $y(x)$ satisfies the differential equation $y' - y \tan x = 2x \sec x$ and $y(0) = 0$, then

(1) $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8\sqrt{2}}$

(2) $y'\left(\frac{\pi}{4}\right) = \frac{\pi^2}{18}$

(3) $y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{9}$

(4) $y'\left(\frac{\pi}{3}\right) = \frac{4\pi}{3} + \frac{2\pi^2}{3\sqrt{3}}$

19. If $y(t)$ is a solution of $(1 + t) \frac{dy}{dt} - ty = 1$ and $y(0) = -1$ then $y(1)$ equals

(1) $-\frac{1}{2}$ (2) $e + \frac{1}{2}$

(3) $e - \frac{1}{2}$ (4) $\frac{1}{2}$



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (1)
2. (2)
3. (4)
4. (1, 2, 3)
5. (3)
6. (4)
7. (2)
8. (3)
9. (1)
10. (4)

11. (1)
12. (4)
13. (1)
14. (2)
15. (1)
16. (1)
17. (3)
18. (1, 4)
19. (1)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>